

This assignment does not need to be handed in.

1. Exercise 1.1 from Tuckman & Serrat
2. Exercise 1.5 from Tuckman & Serrat
3. Exercise 2.1 from Tuckman & Serrat
4. Exercise 2.2 from Tuckman & Serrat
5. A state lottery advertises a jackpot of \$1,000,000. In the fine print it is written that the winner receives 40 annual payments of \$25,000. If the term structure is flat at 6% (i.e., $\hat{r}(t) = .06$ for all $t > 0$), what is the “true value” of the jackpot? (Here “true value” means “net present value”. This is Exercise 3.8 from the second edition of Tuckman.)
6. An investor deposits \$15,000 in a bank today at 6% annual interest. Assuming that no additional deposits are made and that no money is withdrawn, how large will the account balance be 23 years from today if
 - (a) the interest is computed using the simple interest convention?
 - (b) the interest is compounded annually?
 - (c) the interest is compounded quarterly?
 - (d) the interest is compounded monthly?
 - (e) the interest is compounded continuously?(You should assume that the 6% annual interest rate applies to deposits of any length.)
7. A bank is paying interest compounded continuously at the annual rate r . The rate is independent of the length of the deposit and it is known now that the same rate will apply to deposits made in the future. A customer calls the bank and asks: “If I deposit \$1,000 today and an additional \$2,000 6 months from today and make no other deposits or withdrawals, what will my account balance be one year from today?” The bank answers \$3,128.98. Determine r .
8. A zero coupon bond with face value $F = \$5,000$ and maturity $T = 7$ is currently trading at \$3,637.44. Find the spot rate $\hat{r}(7)$ and the effective spot rate $\hat{R}(7)$.
9. Annuities that will pay \$1,000 twice per year for the next 5 years (i.e., 10 payments in all) are currently being sold for \$8,000 per annuity. Coupon bonds having maturity 5 years, face value $F = \$1,000$, and annual coupon rate $q = .1$ are being issued today at the price of \$1,100 per bond. Find the spot rate $\hat{r}(5)$ and the effective spot rate $\hat{R}(5)$.